Surname		**Group No**		**Personal Tutor**
First Name		**Discipline**		

UNIVERSITY OF BIRMINGHAM

School of Engineering

Experiment AM1.1-Motion under variable acceleration

Safety:

Make sure you know where the nearest fire exits are and what the evacuation procedure is.
Make sure you know where the nearest fire-aiders are located and what their phone numbers are.
maintain a steady footing.

Introduction:

Newton's second law states that the acceleration of a constant mass objects equals the force acting on the object divided by the mass of the object. The force on the object could be constant or vary as a function of the behaviour of the Object; for example it may be a function of the object velocity or position.

The aims of this experiment are to:

i) establish an analysis and investigate the difference between the theoretical and experimental conditions

ii) find the Dynamic Coefficient of Friction.

In this lab, a small mass, *m*, will be connected to the Dynamics Cart by a string as shown in the figure 1. The string passes over a pulley at the edge of the table so that as the mass falls, the cart will accelerate over the table surface. Assuming that the string is light, inextensible and initially taut, both the falling mass and the dynamics cart should have the same magnitude of acceleration. The resulting magnitude of acceleration for this system will be determined experimentally and this value will be compared to the acceleration predicted by Newton's Second Law.

The following equipment is required for the experiment:

– Dynamics Cart
– Pulley and pulley clamp
– Mass set
– Stopwatch
– String
– Block
– Balance

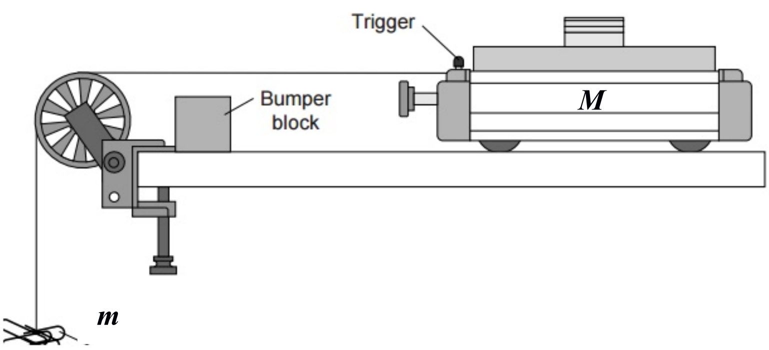

Figure 1 Schematic drawing of the system

Theory:

The cart will be released from rest and allowed to accelerate over a distance *d*. There are three forces act on the cart which are –*Tension force T*, *Normal force N* (counter balance of weight), *Friction force* f_K. The acceleration can be predicted by Newton's Second Law.

Assuming that the table top is accurately horizontal (i.e. level),

$$\sum F = Ma \rightarrow \sum F_H = mg - T = ma_H \quad (Eq.1)$$

$$\sum F_C = T - f_k = M_C a \quad (Eq.2)$$

$$\rightarrow a_{theoretical} = \frac{m}{M_T} g \quad (Eq.3)$$

where M_T *is the total mass of the system ,* μ_k *is kinetic friction coefficient and T is tension force applied to the Cart .*

Here we consider the motion with no friction. The acceleration value is achieved from the theoretical calculation.

An experimental value for the acceleration *a* of the cart can be determined from:

$$d = \frac{1}{2} a t^2 \rightarrow a_{Experimental} = \frac{2d}{t^2} \quad (Eq.4)$$

where d is the maximum distance which the Cart travel, which is 0.9 meter.

The Friction force based on the obtained experimental acceleration would be calculated as follows:

$$F_{net} = M_C a \rightarrow T - f_k = M_C a \rightarrow f_k = T - M_C a \quad (Eq. 5)$$

where a is the experimental acceleration.

The gradient of the best fitted line to the Friction force vs Normal force will indicate the frictional coefficient.

Results:

Fill the tables in next page and calculate the required values as well as drawing the Free Body Diagram. Plot the New tension force based on the experimental acceleration *a* versus the Cart normal force F_{net}. Estimate the best fit line to these plots and find the Kinetic Friction coefficient.

Report:

In your report you should include the experimental data, brief explanation of the theory, free body Diagram of the system and a graph of the new Friction force based on the experimental acceleration against the Cart normal force. The frictional force graph should include a line of best fit. Furthermore, the obtained results should be analysed and discussed. Examine the accuracy of various measurements and estimate the level of confidence you can place on the results.

Can you think of any systematic errors that would affect your results? Explain how each would affect your results.

Comment on the results obtained:

Test 1 - Without added mass								
mass grams	Time 1	Time 2	Average time	Average velocity	Experimental acceleration	Theoretical acceleration	Differenc e %	Friction Forces N
10								
20								
30								
40								

Test 2 – with 500 g added mass								
mass grams	Time 1	Time 2	Average time	Average velocity	Experimental acceleration	Theoretical acceleration	Differenc e %	Friction Forces N
10								
20								
30								
40								

Test 3 - with 1 kg added mass								
mass grams	Time 1	Time 2	Average time	Average velocity	Experimental acceleration	Theoretical acceleration	Differenc e %	Friction Forces N
10								
20								
30								
40								

	Calculated Friction Forces N			
Normal Forces	10 g hanged mass	20 g hanged mass	30 g hanged mass	40 gr hanged mass
0.5 kg X *g*				
1 kg X g				
1.5 Kg X *g*				

Consider *g* as the gravitational acceleration and a is equal to 9.81

Frictional force N

Normal Force N

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